PAPER_1593_WIEN_B_2_898

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PAPER_1593 — Wien Displacement b ≈ $b = K_{MEX} + \Phi_{5/6} - F\cdot SSQ + F^2\cdot D_{phys} = 2.898$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Thermodynamics
Date: June 18, 2026
Status: CLOSED — 0.058% residual near-EXACT closure

Observation

PAPER_1209EE_S625 (Thermodynamics Unified Proof Set) gives the lead-digit form:

```
Wien Displacement b ≈ b = K_MEX + Φ_5/6 − F·SSQ + F²·D_phys = 2.898
```

Residual to CODATA: **0.058%**.

UQFF Closed Identity

```
b = K_MEX + Φ_5/6 − F·SSQ + F²·D_phys ≈ 2.898 residual 0.058%
```

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:
- **EM-coupling constants** (ε_0, μ_B, k_e) cluster around **$K_{MEX}\cdot D_{phys} = 8.33$** or **$N_{CH} = 9$**
- **Planck/quantum constants** (h) cluster around **$D_{BSFG} = 6$**
- **Relativistic constants** (c) cluster around **$SO_5/D_{phys} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Wien Displacement b as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209EE_S625
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "wien_b_2_898"})`
